

ORIGINAL ARTICLE

Mim8 Bispecific Antibody Prophylaxis in Hemophilia A with or without Inhibitors

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ABSTRACT

BACKGROUND

Mim8 (denecimig), a bispecific antibody mimicking activated factor VIII, was developed for bleeding prophylaxis in patients with hemophilia A with or without factor VIII inhibitors.

METHODS

In this phase 3, randomized trial, we assigned patients 12 years of age or older with hemophilia A with or without inhibitors to receive subcutaneous Mim8 once weekly or once monthly at a dose tiered according to body weight and given in a fixed injection volume (0.8 ml). Patients who had been receiving on-demand treatment before the trial were assigned in a 1:1:1 ratio to continue on-demand treatment (group 1) or receive Mim8 once weekly (group 2a) or once monthly (group 2b). Patients who had been receiving clotting factor concentrates during a run-in phase were assigned in a 1:1 ratio to receive Mim8 once weekly (group 3) or once monthly (group 4). The first primary end point was the annualized rate of treated bleeding events (those treated with a coagulation factor product) in an evaluation of Mim8 in group 2a and Mim8 in group 2b as compared with on-demand treatment in group 1. The second was the annualized rate of treated bleeding events in an inpatient evaluation of Mim8 in group 3 and Mim8 in group 4 as compared with clotting factor concentrate prophylaxis during the run-in phase.

RESULTS

Of the 58 patients in the pretrial on-demand treatment cohort, 17 were assigned to group 1, 21 to group 2a, and 20 to group 2b. The estimated mean annualized rate of treated bleeding events was 0.57 (95% confidence interval [CI], 0.25 to 1.30) in group 2a and 0.20 (95% CI, 0.06 to 0.71) in group 2b, as compared with 15.76 (95% CI, 10.70 to 23.20) in group 1 (relative decrease, 96.4% and 98.7%, respectively; $P < 0.001$ for both comparisons). Of the 196 patients in the pretrial prophylaxis cohort, 98 each were assigned to group 3 or group 4. The estimated mean annualized rate of treated bleeding events was 2.25 (95% CI, 1.37 to 3.71) in group 3, as compared with 4.90 (95% CI, 3.65 to 6.56) during the run-in phase (relative decrease, 54.0%; $P = 0.006$), and 1.78 (95% CI, 1.18 to 2.71) in group 4, as compared with 3.12 (95% CI, 2.25 to 4.32) (relative decrease, 42.8%; $P = 0.006$). Injection-site reactions were reported in 103 of 4005 injections (2.6%). No patient was reported to have a thromboembolic event or clinical evidence of neutralizing anti-Mim8 antibodies.

CONCLUSIONS

Among patients with hemophilia A with or without inhibitors, Mim8 prophylaxis was superior to on-demand treatment and clotting factor concentrate prophylaxis regarding the annualized rate of treated bleeding events. (Funded by Novo Nordisk; FRONTIER2 ClinicalTrials.gov number, NCT05053139.)

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HEMOPHILIA A IS AN X-LINKED BLEEDING disorder caused by factor VIII deficiency that results in bleeding, especially into joints, and hemophilic arthropathy.^{1,2} The standard care in patients with hemophilia A is prophylaxis with clotting factor concentrates or activated factor VIII (factor VIIIa) mimetics to prevent bleeding.^{2,3}

Factor VIIIa mimetics that involve less frequent subcutaneous infusions can help reduce the burden of treatment.⁴ These drugs are designed to emulate the cofactor function of endogenous factor VIIIa by bridging factor IXa and factor X to restore thrombin generation.⁵⁻⁹

Mim8 (denecimig) is a factor VIIIa-mimetic bispecific antibody. Mim8 has distinct binding epitopes that are optimized for high cofactor activity and low binding affinity for factor X and factor IXa — features that minimize the risk of nonspecific binding while effectively activating factor X.¹⁰⁻¹³ Mim8 has a thrombin-generation potency that is approximately 15 times as high as that of the factor VIIIa mimetic emicizumab *ex vivo* and that of a sequence-identical analogue of emicizumab *in vitro*.^{11,13} Because of this increased thrombin-generation potency, Mim8 can be given as a one-time loading dose and a fixed, low-volume injection for all dose-administration frequencies. Mim8 can be administered by means of a prefilled device to minimize administration errors and drug waste.^{11,14-18} A new, tiered-dosing approach was implemented in phase 3 trials, with Mim8 administered in a fixed injection volume of 0.8 ml at a dose based on body-weight ranges and the patient's preferred frequency of administration. This tiered approach was designed to remove the need for dose calculations, reduce the burden of treatment administration, and avoid drug wastage. Here, we report week 26 results of FRONTIER2, a phase 3a trial that investigated the efficacy and safety of subcutaneous Mim8 prophylaxis, administered once weekly or once monthly, in patients with hemophilia A of any severity with or without factor VIII inhibitors.

METHODS

TRIAL OVERSIGHT

This phase 3a, prospective, open-label, randomized, controlled trial was designed by Novo Nordisk (trial sponsor). The trial was approved by all the relevant institutional review boards and

independent ethics committees. Written informed consent was obtained before trial-related activities. Data were collected by investigators and staff at the trial sites. Statisticians at Novo Nordisk conducted the analyses. The first draft of the manuscript was written by medical writers, funded by the sponsor, in accordance with Good Publication Practice guidelines. The authors wrote and reviewed subsequent drafts of the manuscript and made the decision to submit the manuscript for publication. The authors vouch for the accuracy and completeness of the data and for the adherence of the trial to the protocol (available with the full text of this article at [NEJM.org](https://www.nejm.org)).

TRIAL DESIGN AND PATIENTS

Patients with hemophilia A with or without inhibitors were assigned to one of five groups during the 26-week main phase of the trial (Fig. 1). Patients who had been receiving on-demand treatment before enrollment (pretrial on-demand treatment cohort) were randomly assigned in a 1:1:1 ratio to continue on-demand treatment (group 1) or to receive Mim8 once weekly (group 2a) or once monthly (group 2b). Randomization was stratified according to the historical annualized rate of bleeding events during the 12 months before screening (≥ 18 events per year vs. < 18 events per year) and factor VIII inhibitor status (with inhibitors vs. without inhibitors). Patients who had been receiving clotting-factor concentrate as prophylaxis before enrollment and then continued clotting factor prophylaxis during a run-in phase lasting 26 to 52 weeks before the 26-week main phase (pretrial prophylaxis cohort) were randomly assigned in a 1:1 ratio to receive Mim8 once weekly (group 3) or once monthly (group 4). Randomization was stratified according to hemophilia severity (severe vs. nonsevere, defined by endogenous factor VIII activity of $< 1\%$ and $\geq 1\%$, respectively) at diagnosis.

The main phase of the trial was followed by a 26-week extension phase; data from the extension phase are not presented here. Patients attended a follow-up visit 21 weeks after the last dose of the trial agent if they had discontinued the trial or had not transferred to the open-label extension trial (FRONTIER4; [ClinicalTrials.gov](https://clinicaltrials.gov) number, NCT05685238) after completion of the 26-week extension phase.

Patients who were 12 years of age or older and had hemophilia A with or without inhibitors



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were eligible for the trial. Patients in the pretrial on-demand treatment cohort needed to have had at least five treated bleeding events during the 26 weeks before screening. Patients in the pretrial prophylaxis cohort who had nonsevere hemophilia A needed to have had at least one treated bleeding event during the 26 weeks before screening. Patients were excluded if they had received prophylaxis with a non-clotting factor product within 6 months or five half-lives of the product before randomization, whichever was shorter. Additional information about the inclusion and exclusion criteria is provided in Section 1, Methods, in the Supplementary Appendix (available at NEJM.org).

TREATMENT

Mim8 was administered subcutaneously once weekly or once monthly in a fixed volume of

0.8 ml. The Mim8 dose was tiered according to body weight (30 to <45 kg or ≥ 45 kg).¹⁸ To rapidly achieve steady-state plasma levels of Mim8, patients weighing 30 to less than 45 kg were given a one-time loading dose of 24 mg if receiving Mim8 once weekly or 40 mg if receiving Mim8 once monthly and those weighing at least 45 kg were given a one-time loading of 55 mg if receiving Mim8 once weekly or 92 mg if receiving Mim8 once monthly. Patients weighing 30 to less than 45 kg then received maintenance doses of 4 mg once weekly or 20 mg once monthly, and those weighing at least 45 kg were then given maintenance doses of 9 mg once weekly or 46 mg once monthly (Table S1 in the Supplementary Appendix). Recombinant factor VIIa was recommended as first-line therapy for breakthrough bleeding in patients with a high titer of factor VIII inhibitors; concomitant use of activated pro-

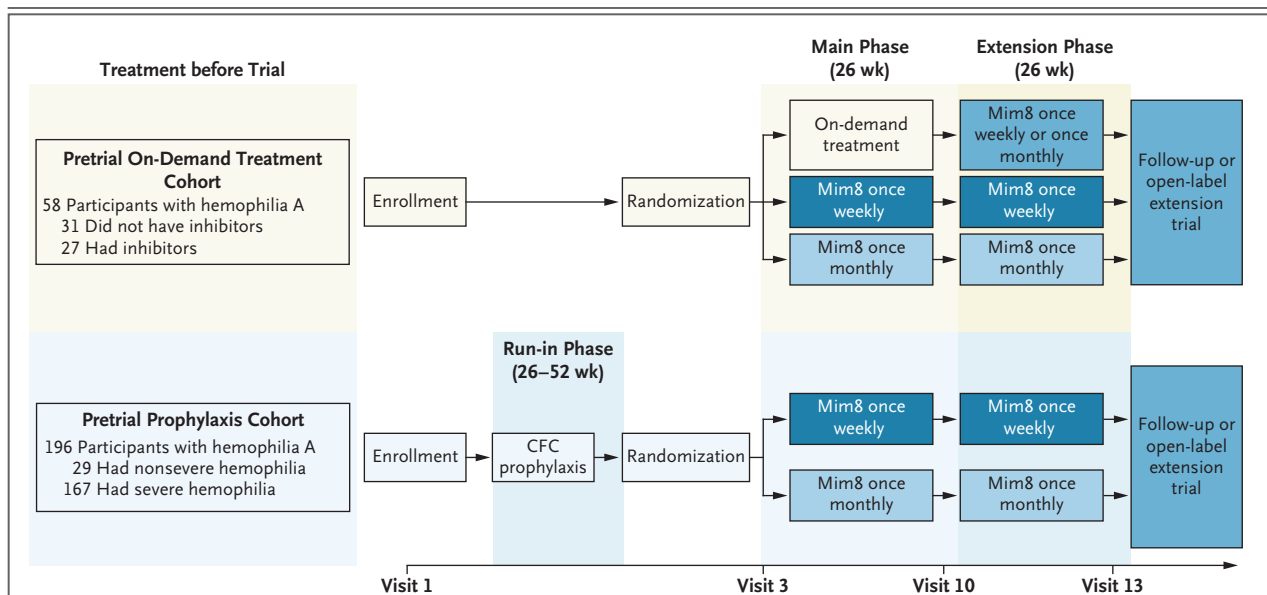


Figure 1. Trial Design.

Patients with hemophilia A with or without factor VIII inhibitors were assigned to one of five groups during the 26-week main phase of the trial. Patients who had been receiving on-demand treatment before enrollment (pretrial on-demand treatment cohort) were randomly assigned in a 1:1:1 ratio to continue on-demand treatment (group 1) or to receive Mim8 prophylaxis once weekly (group 2a) or once monthly (group 2b). Randomization was stratified according to the historical annualized rate of bleeding events during the 12 months before screening (≥ 18 events per year vs. < 18 events per year) and factor VIII inhibitor status (with inhibitors vs. without inhibitors, with the presence of inhibitors defined by use or prescription of bypassing treatment during the 6 months before screening). Patients needed to have had at least five treated bleeding events during the 26 weeks before screening. Patients who had been receiving clotting factor concentrate (CFC) as prophylaxis before enrollment and then continued clotting factor concentrate prophylaxis during a run-in phase lasting 26 to 52 weeks before the trial (pretrial prophylaxis cohort) were randomly assigned in a 1:1 ratio to receive Mim8 prophylaxis once weekly (group 3) or once monthly (group 4). Randomization was stratified according to hemophilia severity (severe vs. nonsevere, defined by endogenous factor VIII activity of $< 1\%$ and $\geq 1\%$, respectively) at diagnosis. Patients with a factor VIII activity of at least 1% needed to have had at least one treated bleeding event during the 26 weeks before screening.

thrombin complex concentrate and Mim8 was discouraged (see Section 1, Methods, in the Supplementary Appendix).

OBJECTIVES AND END POINTS

The primary objective was to assess the efficacy of Mim8 in preventing bleeding. The first primary end point was the annualized rate of treated bleeding events in a comparison of Mim8 once-weekly prophylaxis (group 2a) and Mim8 once-monthly prophylaxis (group 2b) with on-demand treatment (group 1). The second primary end point was the annualized rate of treated bleeding events in an inpatient comparison of Mim8 once-weekly prophylaxis (group 3) and Mim8 once-monthly prophylaxis (group 4) with clotting factor concentrate prophylaxis during the run-in phase. Treated bleeding events were defined as any bleeding event treated with a coagulation factor product.

Secondary end points included bleeding event subtypes (spontaneous, traumatic, joint, and target-joint bleeding events, with target-joint events defined by ≥ 3 spontaneous bleeding events in the same joint during the 6 months before the date of assessment), amount of clotting factor concentrate used, injection-site reactions, occurrence of anti-Mim8 antibodies, and Mim8 concentration. The peak height for thrombin generation was measured with a thrombin-generation assay in platelet-poor plasma triggered with tissue factor. The peak height for thrombin generation indicates the maximum amount of thrombin produced after the initiation and propagation of coagulation; a higher peak height indicates a more efficient coagulation system. Adverse events were coded according to the *Medical Dictionary for Regulatory Activities* (MedDRA), version 26.1. Details regarding estimands and additional information about the trial objectives and end points are provided in Section 1, Methods, in the Supplementary Appendix.

STATISTICAL ANALYSIS

For the analysis of the first primary end point (annualized rate of treated bleeding events in the pretrial on-demand treatment cohort), we calculated that 45 patients would provide a power of 86% to confirm that prophylaxis with Mim8, given once weekly or once monthly, was superior to on-demand treatment. For the analysis of the second primary end point (annualized rate of

treated bleeding events in the pretrial prophylaxis cohort), we determined that 186 patients would provide a power of more than 99% to confirm that prophylaxis with Mim8, given once weekly or once monthly, was noninferior to prophylaxis with clotting factor concentrate.

The full analysis population included all the patients who had undergone randomization. The safety analysis population included all the patients who had undergone randomization and received at least one dose of Mim8. The primary efficacy analysis was conducted on an intention-to-treat basis and included all the patients in the full analysis population. All end-point events that occurred after randomization were included in the primary efficacy analysis, irrespective of treatment discontinuation, temporary treatment interruption, or concomitant use of clotting factor concentrate (treatment-policy approach). End-point events that occurred during the main phase were included in the analysis of the first primary end point; those that occurred during the run-in phase and the main phase were included in the analysis of the second primary end point. The 26-week main phase could include a follow-up period after treatment discontinuation. Details regarding the treatment strategies used to handle intercurrent events in the intention-to-treat analysis and the hypothetical strategies used to handle intercurrent events (except for minor surgery) in the prespecified primary analysis are presented in Section 1, Methods, and Section 2 in the Supplementary Appendix.

Primary end points and bleeding event subtypes were analyzed with the use of a negative binomial model, with the logarithm of exposure time as an offset variable. Ratios of annualized rates of treated bleeding events are provided with 95% confidence intervals and P values.

Safety assessments were conducted in the safety analysis population and are reported with descriptive statistics. Injection-site reactions and the occurrence of anti-Mim8 antibodies were summarized descriptively, without formal statistical testing. Remaining secondary end points were evaluated in the full analysis population with descriptive statistics.

No data for the primary end points were missing, and missing covariates were not used as stratification factors during randomization. We performed prespecified sensitivity analyses to assess the robustness of the primary analysis results.

The type I error for the primary end points was controlled for multiplicity with the use of a hierarchical testing procedure (Fig. S1). In the pretrial on-demand treatment group, we compared the annualized rate of treated bleeding events to assess the superiority of Mim8 once-weekly prophylaxis (group 2a) to on-demand treatment (group 1). If this comparison met the criterion for superiority (defined as an upper limit of <1 for the 95% confidence interval of the ratio of the annualized rate of treated bleeding events), we assessed the superiority of Mim8 once-monthly prophylaxis (group 2b) as compared with on-demand treatment (group 1).

In the pretrial prophylaxis group, we first assessed the noninferiority of Mim8 once-weekly prophylaxis (group 3), as compared with clotting factor concentrate prophylaxis, regarding the annualized rate of treated bleeding events. The noninferiority margin was 2.0 if the annualized rate of treatment bleeding events was 3 or less; otherwise, the noninferiority margin was 1.5. We then assessed the noninferiority of Mim8 once-monthly prophylaxis (group 4) as compared with clotting factor concentrate prophylaxis. If noninferiority was shown in both comparisons, superiority was assessed in the same sequence. Comparisons conducted outside these hierarchical analyses were exploratory and are reported with 95% confidence intervals only; these confidence intervals were not adjusted for multiple comparisons and may not be used to infer definitive treatment effects. Additional details about the statistical analyses are provided in Section 1, Methods, and Section 2 in the Supplementary Appendix.

RESULTS

TRIAL POPULATION

Overall, 254 patients were assigned to one of five trial groups. The pretrial on-demand treatment cohort included 58 patients, of whom 17 were assigned to continue on-demand treatment (group 1), 21 to receive Mim8 once-weekly prophylaxis (group 2a), and 20 to receive Mim8 once-monthly prophylaxis (group 2b). The pretrial prophylaxis cohort included 196 patients, of whom 98 each were assigned to receive prophylaxis with Mim8 once weekly (group 3) or once monthly (group 4). Historical annualized rates of bleeding events during the 12 months before screening are reported in Table S2. The main phase of the trial was com-

pleted by 246 patients (Fig. S2). Of the patients who underwent randomization, 4 (2%) were female, 66 (26%) were adolescents (12 to 17 years of age), 212 (84%) had severe hemophilia A, 31 (12%) had factor VIII inhibitors, and 242 (95%) weighed at least 45 kg (Table 1). Eight patients (3%) discontinued treatment — 3 because of adverse events, 1 because of a lack of efficacy as judged by the patient, and 4 because of other reasons.

EFFICACY END POINTS

In the intention-to-treat analysis of the pretrial on-demand treatment cohort, the estimated mean annualized rate of treated bleeding events was 0.57 (95% confidence interval [CI], 0.25 to 1.30) with Mim8 once-weekly prophylaxis (group 2a) and 0.20 (95% CI, 0.06 to 0.71) with Mim8 once-monthly prophylaxis (group 2b), as compared with 15.76 (95% CI, 10.70 to 23.20) with on-demand treatment (group 1) (Table 2). Mim8 prophylaxis was superior to on-demand treatment for the first primary end point, with a relative reduction in the estimated annualized rate of treated bleeding events of 96.4% with the once-weekly regimen (rate ratio, 0.04; 95% CI, 0.01 to 0.09; $P<0.001$) and 98.7% with the once-monthly regimen (rate ratio, 0.01; 95% CI, <0.01 to 0.05; $P<0.001$).

In the intention-to-treat analysis of the pretrial prophylaxis cohort, the estimated mean annualized rate of treated bleeding events was 2.25 (95% CI, 1.37 to 3.71) with Mim8 once-weekly prophylaxis (group 3), as compared with 4.90 (95% CI, 3.65 to 6.56) with clotting factor concentrate prophylaxis, and 1.78 (95% CI, 1.18 to 2.71) with Mim8 once-monthly prophylaxis (group 4), as compared with 3.12 (95% CI, 2.25 to 4.32) with clotting factor concentrate prophylaxis (Table 2). The relative reduction in the annualized rate of treated bleeding events as compared with clotting factor concentrate prophylaxis was 54.0% with the once-weekly regimen (rate ratio, 0.46; 95% CI, 0.26 to 0.80; $P=0.006$) and 42.8% with the once-monthly regimen (rate ratio, 0.57; 95% CI, 0.38 to 0.85; $P=0.006$), showing that Mim8 prophylaxis, given once weekly or once monthly, was superior to clotting factor concentrate prophylaxis for the second primary end point. Results of the prespecified efficacy analyses in both cohorts are shown in Section 2 of the Supplementary Appendix.

Figure S3 shows ratios of estimated mean an-

Table 1. Baseline Characteristics of the Patients.*

Characteristic	Pretrial On-Demand Treatment			Pretrial Prophylaxis	
	On-Demand Treatment (N=17)	Mim8 Once Weekly (N=21)	Mim8 Once Monthly (N=20)	Mim8 Once Weekly (N=98)	Mim8 Once Monthly (N=98)
Age					
Mean (range) — yr	31 (13–52)	32 (13–64)	34 (13–64)	33 (12–68)	32 (12–69)
Distribution — no. (%)					
12–17 yr	5 (29)	5 (24)	4 (20)	26 (27)	26 (27)
≥18 yr	12 (71)	16 (76)	16 (80)	72 (73)	72 (73)
Sex — no. (%)					
Female	1 (6)	0	2 (10)	0	1 (1)
Male	16 (94)	21 (100)	18 (90)	98 (100)	97 (99)
Race or ethnic group — no. (%)†					
Asian	11 (65)	11 (52)	11 (55)	12 (12)	17 (17)
Black	1 (6)	1 (5)	2 (10)	1 (1)	2 (2)
Native Hawaiian or other Pacific Islander	0	0	0	1 (1)	0
White	4 (24)	9 (43)	7 (35)	72 (73)	68 (69)
Other	1 (6)	0	0	7 (7)	7 (7)
Body weight					
Overall — kg	63.3±16.1	72.0±20.9	69.5±18.9	76.7±20.3	78.2±19.7
Distribution — no. (%)					
30 to <45 kg	1 (6)	2 (10)	4 (20)	5 (5)	0
≥45 kg	16 (94)	19 (90)	16 (80)	93 (95)	98 (100)
Severity of hemophilia — no. (%)					
Mild	2 (12)	2 (10)	1 (5)	0	1 (1)
Moderate	2 (12)	5 (24)	1 (5)	14 (14)	14 (14)
Severe	13 (76)	14 (67)	18 (90)	84 (86)	83 (85)
Current inhibitor status — no. (%)					
Positive	8 (47)	10 (48)	9 (45)	2 (2)	2 (2)
Negative	9 (53)	11 (52)	11 (55)	96 (98)	96 (98)

* Plus–minus values are means ±SD. Patients with hemophilia A with or without factor VIII inhibitors were assigned to one of five groups during the 26-week main phase of the trial. Patients who had been receiving on-demand treatment before enrollment (pretrial on-demand treatment cohort) were randomly assigned in a 1:1:1 ratio to continue on-demand treatment or to receive Mim8 prophylaxis once weekly or once monthly. Patients who had been receiving clotting factor concentrate as prophylaxis before enrollment and then continued clotting factor concentrate prophylaxis during a run-in phase lasting 26 to 52 weeks before the trial (pretrial prophylaxis cohort) were randomly assigned in a 1:1 ratio to receive Mim8 prophylaxis once weekly or once monthly. Percentages may not sum to 100 because of rounding.

† Race and ethnic group were reported by the patients. In the pretrial prophylaxis cohort, data were missing for five patients (5%) in the Mim8 once-weekly group and four patients (4%) in the Mim8 once-monthly group.

nualized rates of treated bleeding events in the pretrial on-demand treatment cohort (Mim8 once-weekly prophylaxis and Mim8 once-monthly prophylaxis vs. on-demand treatment) and the pretrial prophylaxis cohort (Mim8 once-weekly prophylaxis and Mim8 once-monthly prophylaxis vs.

clotting factor concentrate prophylaxis) according to bleeding event subtype. In the pretrial on-demand treatment cohort, the relative reduction in the estimated mean annualized rate of treated joint bleeding events as compared with on-demand treatment was 95.0% (rate ratio, 0.05; 95% CI, 0.02

Table 2. Treated Bleeding Events (Intention-to-Treat Analysis).*

Variable	Pretrial On-Demand Treatment				Pretrial Prophylaxis			
	On-Demand Treatment (N=17)	Mim8 Once Weekly (N=21)	Mim8 Once Monthly (N=20)		Run-in Phase	Main Phase	Run-in Phase	Main Phase
Mean exposure time — wk	25.7	25.6	25.8		40.9	25.3	39.1	26.0
Treated bleeding events — no.	146	8	3		383	103	233	88
Mean estimated ABR (95% CI) †	15.76 (10.70–23.20)	0.57 (0.25–1.30)	0.20 (0.06–0.71)		4.90 (3.65–6.56)	2.25 (1.37–3.71)	3.12 (2.25–4.32)	1.78 (1.18–2.71)
Ratio of mean estimated ABRs (95% CI) †‡	—	0.04 (0.01–0.09)	0.01 (<0.01–0.05)		—	0.46 (0.26–0.80)	—	0.57 (0.38–0.85)
Percentage reduction in mean estimated ABR (95% CI) †§	—	96.4 (91.0–98.6)	98.7 (95.2–99.7)		—	54.0 (19.6–73.6)	—	42.8 (14.7–61.6)
P value¶	<0.001	<0.001	<0.001			0.006		0.006
Median estimated ABR (IQR) †	12.24 (8.03–18.06)	0 (0–0)	0 (0–0)		2.11 (0.00–5.72)	0 (0–2.01)	1.49 (0–3.87)	0 (0–2.01)
Distribution of treated bleeding events — no. (%)								
0 events	0	17 (81)	19 (95)		32 (33)	65 (66)	36 (37)	63 (64)
1 or 2 events	2 (12)	3 (14)	0		26 (27)	21 (21)	34 (35)	25 (26)
≥3 events	15 (88)	1 (5)	1 (5)		40 (41)	12 (12)	28 (29)	10 (10)
Severe bleeding — no./total no. of bleeding events (%)	7/146 (4.8)	1/8 (12.5)	0/3		0/383	1/103 (1.0)	2/233 (0.9)	1/88 (1.1)
Treated bleeding events according to type								
No./total no. of bleeding events (%)**								
Spontaneous bleeding	108/146 (74.0)	2/8 (25.0)	3/3 (100)		221/383 (57.7)	59/103 (57.3)	136/233 (58.4)	40/88 (45.5)
Traumatic bleeding	35/146 (24.0)	6/8 (75.0)	0/3		156/383 (40.7)	42/103 (40.8)	96/233 (41.2)	43/88 (48.9)
Surgical bleeding	3/146 (2.1)	0/8	0/3		3/383 (0.8)	2/103 (1.9)	1/233 (0.4)	5/88 (5.7)
Mean estimated ABR (95% CI) †								
Spontaneous bleeding	11.77 (7.48–18.53)	0.17 (0.04–0.74)	0.24 (0.07–0.84)		2.75 (1.96–3.85)	1.26 (0.61–2.63)	1.83 (1.10–3.03)	0.76 (0.43–1.34)
Traumatic bleeding	1.86 (0.87–4.00)	0.22 (0.07–0.70)	0		2.02 (1.45–2.83)	0.89 (0.58–1.37)	1.30 (0.94–1.80)	0.87 (0.56–1.35)
Bleeding into joints	10.57 (6.34–17.63)	0.53 (0.21–1.34)	0		3.68 (2.70–5.02)	1.62 (0.90–2.93)	2.07 (1.44–2.97)	1.10 (0.61–1.99)
Recurrent bleeding into joints	3.85 (1.99–7.47)	0.30 (0.09–1.01)	0		1.12 (0.62–2.02)	0.38 (0.13–1.06)	0.48 (0.21–1.10)	0.34 (0.11–1.08)
Treated bleeding events according to location — no./total no. of bleeding events (%)								
Joints								

Overall	97/146 (66.4)	7/8 (87.5)	0/3	294/383 (76.8)	75/103 (72.8)	155/233 (66.5)	55/88 (62.5)
Spontaneous bleeding	81/97 (83.5)	2/7 (28.6)	0/3	178/294 (60.5)	50/75 (66.7)	102/155 (65.8)	29/55 (52.7)
Other	49/146 (33.6)	1/8 (12.5)	3/3 (100)	89/383 (23.2)	28/103 (27.2)	78/233 (33.5)	33/88 (37.5)

* The intention-to-treat analysis included all the patients who had undergone randomization. Percentages may not sum to 100 because of rounding. IQR denotes interquartile range.

† The annualized rate of treated bleeding events (ABR) was estimated with a negative binomial regression model.

‡ Shown are ratios of the annualized rate of treated bleeding events (Mim8 prophylaxis vs. on-demand treatment) among patients in the pretrial on-demand treatment cohort and ratios of the annualized rate of treated bleeding events (Mim8 prophylaxis during the main phase vs. clotting factor concentrate prophylaxis during the run-in phase) among patients in the pretrial prophylaxis cohort.

§ Shown are percentage reductions in the annualized rate of treated bleeding events (Mim8 prophylaxis vs. on-demand treatment) among patients in the pretrial on-demand treatment cohort and percentage reductions in the annualized rate of treated bleeding events (Mim8 prophylaxis during the main phase vs. clotting factor concentrate prophylaxis during the run-in phase) among patients in the pretrial prophylaxis cohort.

¶ P values are for comparisons of the annualized rate of treated bleeding events (Mim8 prophylaxis vs. on-demand treatment) among patients in the pretrial on-demand treatment group and comparisons of the annualized rate of treated bleeding events (Mim8 prophylaxis during the main phase vs. clotting factor concentrate prophylaxis during the run-in phase) among patients in the pretrial prophylaxis cohort.

|| Severe bleeding events included intracranial, retroperitoneal, iliopsoas, and internal neck bleeding; muscle bleeding with compartment syndrome; bleeding associated with a clinically significant decrease in hemoglobin level (>3 g per deciliter); bleeding that resulted in hospitalization; and life-threatening bleeding.

** In the pretrial prophylaxis cohort, data from the run-in phase were missing for three treated bleeding events (0.8%) in the Mim8 once-weekly group.

to 0.14) with Mim8 once-weekly prophylaxis; no treated joint bleeding events were observed with Mim8 once-monthly prophylaxis. In the pretrial prophylaxis cohort, the relative reduction as compared with clotting factor concentrate prophylaxis was 55.9% with Mim8 once-weekly prophylaxis (rate ratio, 0.44; 95% CI, 0.23 to 0.83) and 46.6% with Mim8 once-monthly prophylaxis (rate ratio, 0.53; 95% CI, 0.32 to 0.90). Details regarding untreated bleeding events are shown in Tables S3 and S4.

In the pretrial on-demand treatment cohort, none of the patients continuing on-demand treatment, 81% of those receiving Mim8 once-weekly prophylaxis, and 95% of those receiving Mim8 once-monthly prophylaxis reported zero treated bleeding events (Table 2). The estimated mean annualized rate of treated bleeding events with Mim8 once weekly and Mim8 once monthly combined was 0.09 among patients with inhibitors and 0.78 among those without inhibitors (Fig. S4A).

In the pretrial prophylaxis cohort, 66% of the patients receiving Mim8 once weekly and 64% of those receiving Mim8 once monthly reported zero treated bleeding events (Table 2). One patient who received Mim8 once weekly for 108 days reported 22 bleeding events. All 22 events were considered by the investigator to be spontaneous and mild to moderate in severity; 19 events were in one joint (ankle), and the patient had no anti-Mim8 antibodies. In a post hoc sensitivity analysis of Mim8 once-weekly prophylaxis that excluded this patient, the estimated mean annualized rate of treated bleeding events was lower than in the primary analysis (1.80 vs. 2.25), with a higher relative reduction as compared with clotting factor concentrate prophylaxis (63.0% vs. 54.0%). The estimated mean annualized rate of treated bleeding events with the Mim8 regimens combined was similar in patients with severe hemophilia and those with nonsevere hemophilia (1.99 and 2.05, respectively) (Fig. S4B).

All but one of the bleeding events that occurred during receipt of Mim8 were treated with factor VIII injections; one patient in the pretrial on-demand treatment cohort received an injection of recombinant factor VIIa (91.3 μ g per kilogram of body weight). In the pretrial on-demand treatment cohort, most bleeding events (9 of 11; 82%) that occurred during receipt of Mim8 were treated with 1 or 2 injections. Across the 10 events in this cohort that were treated with factor VIII,

total doses ranged from 22.8 to 1018.2 IU per kilogram; the total dose of 1018.2 IU per kilogram corresponded to 26 injections and was administered for a single traumatic bleeding event that occurred 3 months after discontinuation of Mim8. In the pretrial prophylaxis cohort, 165 of 191 bleeding events (86%) were treated with 1 or 2 injections of factor VIII and 26 (14%) with 3 or more injections (Fig. S5); total doses of factor VIII ranged from 13.8 to 371.3 IU per kilogram.

SAFETY OUTCOMES

Overall, 486 adverse events were reported with Mim8 during the 26-week main phase of the trial; most events (99%) were mild to moderate in severity (Table 3). No clustering of adverse events according to MedDRA preferred term or system organ class was observed.

In total, 23 of 237 patients (10%) receiving Mim8 reported injection-site reactions; these reactions were observed in 103 of 4005 injections (2.6%). The most common signs and symptoms of injection-site reactions included erythema and pruritus (itching), and one injection-site reaction included burning as a symptom. Most injection-site reactions were transient; all were nonserious and mild in severity (Table 3 and Table S5).

Three patients discontinued Mim8 owing to adverse events; one patient had paralysis of the sixth nerve (assessed by the investigator as probably related to Mim8), one had coronary artery disease (assessed as unrelated to Mim8), and one had sleep disorder and dyspepsia (both assessed as possibly related to Mim8) (Table 3 and Table S6). Serious adverse events are reported in Table S7.

No fatal events, thromboembolic events, or hypersensitivity reactions were observed with Mim8. Anti-Mim8 antibodies were detected during the treatment period in 18 of 254 patients (7%), with no clinical evidence of a neutralizing effect, preservation of the Mim8 concentration and activity, an activated partial-thromboplastin time within the reference range, and no effect on bleeding patterns (Table S8). Results of laboratory tests were as expected, including assessments of coagulation markers (levels of D-dimer, prothrombin fragment 1+2, factor IX antigen, factor X antigen, and fibrinogen) (Fig. S6). No major surgeries were performed; 3 minor surgeries occurred during continued on-demand treatment (in group 1), 11 during receipt of Mim8 once

weekly (in group 3), and 15 during receipt of Mim8 once monthly (in group 4).

PHARMACOKINETIC AND PHARMACODYNAMIC FEATURES

Mean estimated plasma concentrations of Mim8 at steady state were generally similar with the once-weekly and once-monthly regimens. Mean estimated log-transformed trough plasma concentrations during the receipt of Mim8 once weekly were 4.05 and 4.13 μg per milliliter in groups 2a and 3, respectively; mean estimated log-transformed trough plasma concentrations during the receipt of Mim8 once monthly were 3.79 and 4.55 μg per milliliter in groups 2b and 4, respectively (Table S9). Mean estimated log-transformed peak concentrations were 5.12 and 5.22 μg per milliliter in groups 2a and 3, respectively, with Mim8 once weekly and 4.79 and 5.75 μg per milliliter in groups 2b and 4, respectively, with Mim8 once monthly. Arithmetic mean plasma concentrations of Mim8 are reported in Figure 2.

In an exploratory analysis with the use of a thrombin-generation assay, the estimated mean peak height for thrombin generation increased to 39.7 nmol per liter (95% CI, 37.3 to 42.3 [normal range, 27.2 to 152.0 nmol per liter]) when the Mim8 concentration was at steady state. Peak heights for thrombin generation were similar with the once-weekly and once-monthly Mim8 regimens.

DISCUSSION

The current trial showed that Mim8 prophylaxis, administered according to a tiered-dosing approach, was superior to on-demand treatment and clotting factor concentrate prophylaxis regarding the annualized rate of treated bleeding events (primary end point) among patients with hemophilia A with or without inhibitors. During 26 weeks of treatment, 64 to 95% of the patients receiving Mim8 remained free of treated bleeding events. Mim8 led to injection-site reactions in 10% of the patients (2.6% of the injections); no fatal events, thromboembolic events, or hypersensitivity reactions were observed.

In the pretrial prophylaxis cohort, the annualized rates of treated bleeding events with clotting factor concentrate prophylaxis were similar to those in other studies of standard and ex-

tended half-life clotting factor concentrates.¹⁹⁻²¹ Both Mim8 regimens were superior to clotting factor concentrate prophylaxis for the primary end point; this finding indicates that Mim8 may enhance efficacy beyond that of current standard care.^{2,3} After the main phase of the current trial, patients entered a 26-week extension phase and subsequently the phase 3 extension trial for the assessment of long-term outcomes.

Mim8 has been shown to have a thrombin-generation potency that is approximately 15 times as high as that of emicizumab *ex vivo* and *in vitro*.^{11,17} The FRONTIER1 trial showed a clear concentration–response relationship between the Mim8 concentration and the peak thrombin level.¹¹ In the current trial, peak thrombin levels increased into the normal range after the initiation of Mim8 prophylaxis, which is consistent with restoration of thrombin-generation potency. These observations can be attributed to structural and mechanistic differences between Mim8 and emicizumab. Mim8 was optimized to engage an epitope on factor IXa that is located within the serine protease domain, near the active site. Mim8 binds to a factor X epitope at the interface between the serine protease domain and the EGF-2 domain, which induces an increase in the allosteric activation of factor IXa by a factor of more than 20,000. In contrast, emicizumab binds to factor IXa and factor X without inducing substantial allosteric activation, relying primarily on a bridging interaction between the two factors.^{13,22} Without a head-to-head trial, direct comparison with emicizumab is not possible. However, the percentage of patients who had zero treated bleeding events during Mim8 prophylaxis in the current trial was similar to the percentage of patients with or without inhibitors who had zero treated bleeding events during 24 weeks of emicizumab prophylaxis.^{4,19} The low median estimated annualized rates of treated bleeding events observed with Mim8 across all treatment groups were consistent with rates for other non-coagulation factor therapies and extended half-life clotting factor concentrates.^{19,23-25}

The annualized rate of bleeding events is influenced by the recognition of bleeding and prone to subjective assessment and heterogeneity across studies.²⁶ In the current trial, the mean annualized rate of bleeding events with Mim8 in the pretrial on-demand treatment cohort was numeri-

cally lower than in the pretrial prophylaxis cohort. This finding may have been partially influenced by treatment history. During the 12 months before screening, the annualized rate of bleeding events was higher among patients in the pretrial on-demand treatment cohort than among those in the pretrial prophylaxis cohort. Patients in the pretrial on-demand treatment cohort may therefore have had a greater ability during the main phase than those in the pretrial prophylaxis cohort to differentiate between pain due to bleeding and pain due to other causes (e.g., arthropathy and impaired joint health). As a result, misinterpretation of pain symptoms as a bleeding event may have been more common in the pretrial prophylaxis cohort during the main phase, leading to increased reporting of bleeding events. Although these hypotheses lack supporting data, similar observations were made in a phase 2 trial of concizumab.²⁷ Greater reductions in the annualized rate of bleeding events among patients who had previously received on-demand therapy than among those who had previously received prophylaxis have been observed with other non-factor-replacement therapies (e.g., emicizumab)¹⁹ and the recombinant factor VIII replacement therapy efanesoctocog alfa.²⁴ These findings may reflect the greater absolute burden of bleeding at baseline in the on-demand treatment cohorts than in the prophylaxis cohorts, allowing greater potential for reducing the risk of bleeding with prophylaxis.

Few patients discontinued Mim8 because of adverse events; no thromboembolic events, hypersensitivity reactions, or clinical evidence of neutralizing anti-Mim8 antibodies were observed. Safety outcomes were consistent with long-term data from the phase 1–2 FRONTIER1 extension trial.²⁸ The incidence of injection-site reactions with Mim8 was lower than with other subcutaneous agents,²⁹ potentially because of the low injection volume in the current trial and differences in drug formulations and excipients.³⁰⁻³²

Mim8 plasma concentrations with the weekly and monthly regimens were within the expected range reported in dose-finding studies.¹¹ The efficacy of Mim8 once weekly was similar to that of Mim8 once monthly, and plasma concentrations of Mim8 with each regimen were stable over time. These findings indicate that the use of tiered dosing with Mim8 is effective and pro-

Table 3. Adverse Events.*

Variable	Pretrial On-Demand Treatment				Pretrial Prophylaxis†			
	On-Demand Treatment (N = 17)	Mim8 Once Weekly (N = 21)	Mim8 Once Monthly (N = 20)	Mim8 Once Weekly (N = 98)	Run-in Phase	Main Phase	Run-in Phase	Main Phase
Total exposure time — yr	8.4	9.9	9.9	9.9	77.1	46.6	73.8	48.7
Any adverse event‡								
No. of patients (%)	5 (29)	10 (48)	6 (30)	58 (59)	51 (52)	71 (72)	51 (52)	65 (66)
No. of events per total exposure time	1.31	1.51	1.91	1.58	1.57	5.28	1.57	4.23
Serious adverse event								
No. of patients (%)	1 (6)	1 (5)	0	6 (6)	2 (2)	4 (4)	2 (2)	4 (4)
No. of events per total exposure time	0.12	0.10	0	0.10	0.03	0.09	0.03	0.10
Adverse event of special interest								
Thromboembolic event — no. (%)	0	0	0	0	0	0	0	0
Severity of adverse event								
Mild								
No. of patients (%)	3 (18)	7 (33)	6 (30)	49 (50)	48 (49)	67 (68)	48 (49)	61 (62)
No. of events per total exposure time	0.59	1.11	1.91	1.14	1.23	4.44	1.23	3.39
Moderate								
No. of patients (%)	2 (12)	2 (10)	0	20 (20)	17 (17)	24 (24)	17 (17)	21 (21)
No. of events per total exposure time	0.59	0.20	0	0.42	0.31‡	0.79	0.31‡	0.78
Severe								
No. of patients (%)	1 (6)	2 (10)	0	1 (1)	2 (2)	2 (2)	2 (2)	3 (3)
No. of events per total exposure time	0.12	0.20	0	0.03	0.03	0.04	0.03	0.06
Adverse event considered by the investigator as related to Mim8§								
No. of patients (%)	NA	4 (19)	1 (5)	NA	NA	20 (20)	NA	17 (17)
No. of events per total exposure time	NA	0.40	0.20	NA	NA	2.06	NA	0.62

Adverse event considered by the investigator as leading to permanent discontinuation of Mim8¶						
No. of patients (%)	NA	1 (5)	0	NA	2 (2)	NA
No. of events per total exposure time	NA	0.10	0	NA	0.06	NA
Adverse event considered by the investigator as probably or possibly causally related to Mim8§						
Injection-site reaction						
No. of patients (%)	NA	2 (10)	1 (5)	NA	12 (12)	NA
No. of events per total exposure time	NA	0.20	0.20	NA	1.67	NA
Hypersensitivity reaction — no. (%)	NA	0	0	NA	0	NA
Adverse event due to medication error or misuse or abuse of Mim8 — no. (%)						
No. of patients (%)	NA	0	0	NA	1 (1)	NA
No. of events per total exposure time	NA	0	0	NA	0.02	NA

* NA denotes not applicable.

† Visits were scheduled to occur every 12 weeks during the run-in phase and on weeks 1, 2, 4, 8, 16, 21, and 26 during the main phase.

‡ The full list of serious adverse events is provided in Table S7.

§ Decisions by the investigator were made according to the trial protocol.

¶ In the pretrial on-demand treatment cohort, one patient who was receiving Mim8 once weekly discontinued treatment owing to a serious adverse event (sixth nerve paralysis) that was considered by the investigator as probably related to Mim8. In the pretrial prophylaxis cohort, two patients who were receiving Mim8 once monthly discontinued treatment; one discontinued owing to coronary artery disease that was considered by the investigator as probably unrelated to Mim8, and one discontinued owing to sleep disorder and dyspepsia that were considered by the investigator as possibly related to Mim8.

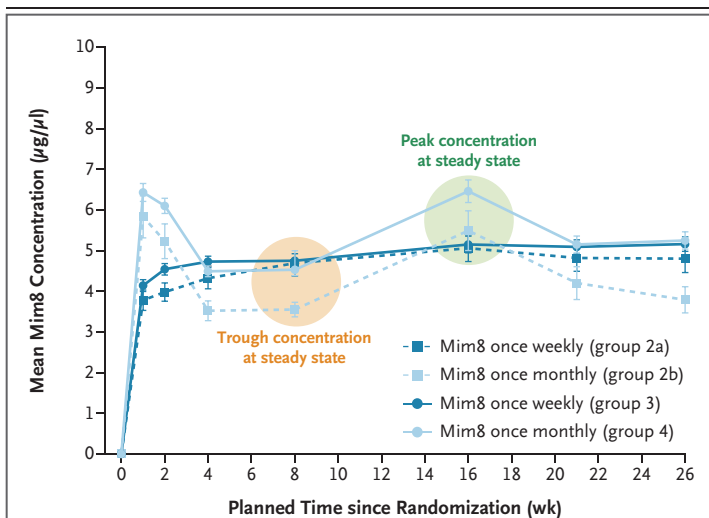


Figure 2. Arithmetic Plasma Concentration of Mim8.

As prespecified in the protocol, the trough concentration of Mim8 at steady state was defined as the estimated plasma concentration at week 8 and the peak concentration of Mim8 at steady state as the estimated plasma concentration at week 16. Concentrations are shown on a linear scale. The planned time since randomization indicates the time of the planned follow-up visit; some visits may not have occurred on the planned day. The I bars indicate standard errors.

vides a flexible and low-frequency approach to dose administration that could improve adherence and reduce the treatment burden.^{33,34} The tiered-dosing approach was informed by pharmacokinetic data from FRONTIER1 and model-based simulations¹⁸ to define a Mim8 target exposure range and appropriate body-weight ranges and doses. The target exposure range guided Mim8 modeling and the creation of tiered doses and is not intended for clinical use. No safety concerns were observed across this exposure range. By removing the need for calculation and adjustment of doses on the basis of body weight, tiered dosing avoids drug wastage. Furthermore, the use of prefilled devices with a low infusion volume reduces the time needed for drug preparation and infusion. These features are advantageous for patients, caregivers, physicians, and payers.³⁵ Although tiered dosing is not individualized according to pharmacokinetic features, it was prospectively optimized by means of pharmacokinetic and pharmacodynamic modeling and early trial data to achieve exposure and response objectives at the population level, even with fluctuating body weights.¹⁸ Further details about the use of mod-

eling for tiered dosing have been described previously.¹⁸

Our trial had limitations. The trial had an open-label design, treated bleeding events were reported by the patients, and the proportion of patients with inhibitors who were enrolled in the pretrial prophylaxis cohort was small owing to the exclusion of patients who were receiving emicizumab.

Mim8 represents a prophylactic subcutaneous treatment for persons at least 12 years of age who are living with hemophilia A, regardless of hemophilia severity and inhibitor status. In the current trial, Mim8 prophylaxis was superior to on-demand treatment and clotting factor concentrate prophylaxis with respect to the annualized rate of treated bleeding events and did not lead to fatal events, thromboembolic events, or hypersensitivity reactions.

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